

Algorithm, Pseudo code and Flowchart

Algorithm

It is a **step-by-step procedure** written in simple and clear language to solve a problem or perform a task.

- ✓ Uses plain English
- ✓ No strict format
- ✓ Easy to understand

Pseudo code

It is a **structured, code-like representation** of an algorithm that uses programming-style keywords but is not written in any specific programming language.

- ✓ Looks like a program
- ✓ Uses keywords like IF, WHILE, FOR
- ✓ Helps in writing actual code

Flowchart

It is a **graphical (diagrammatic) representation** of an algorithm or process, using **symbols and arrows** to show the sequence of steps.

SYMBOLS

start/stop

input/output

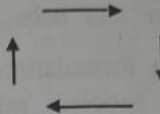
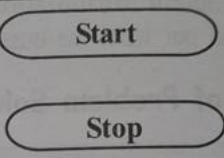
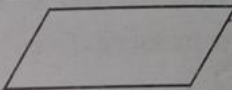
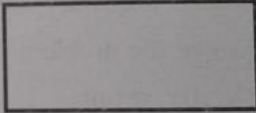
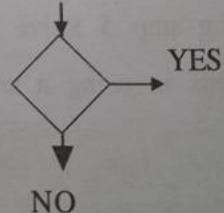
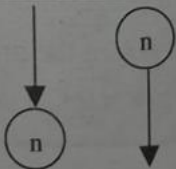
process

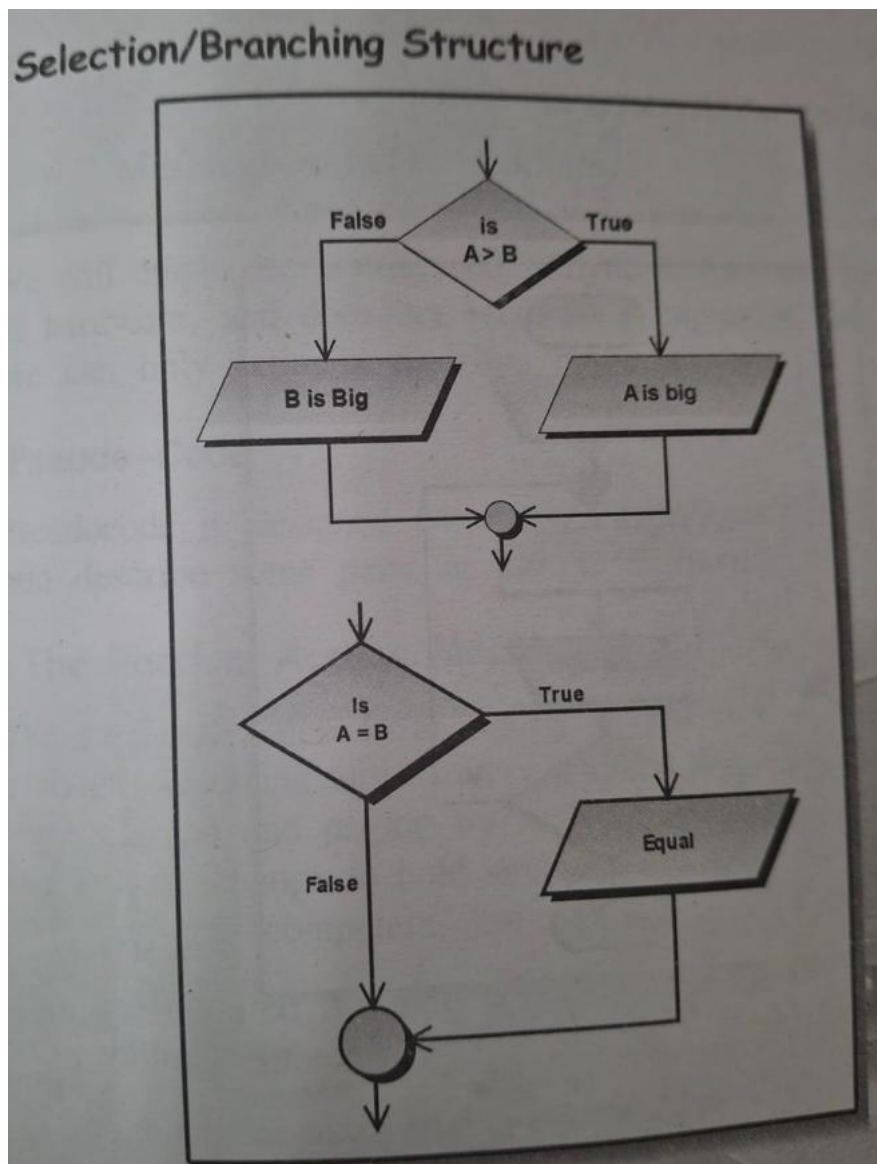
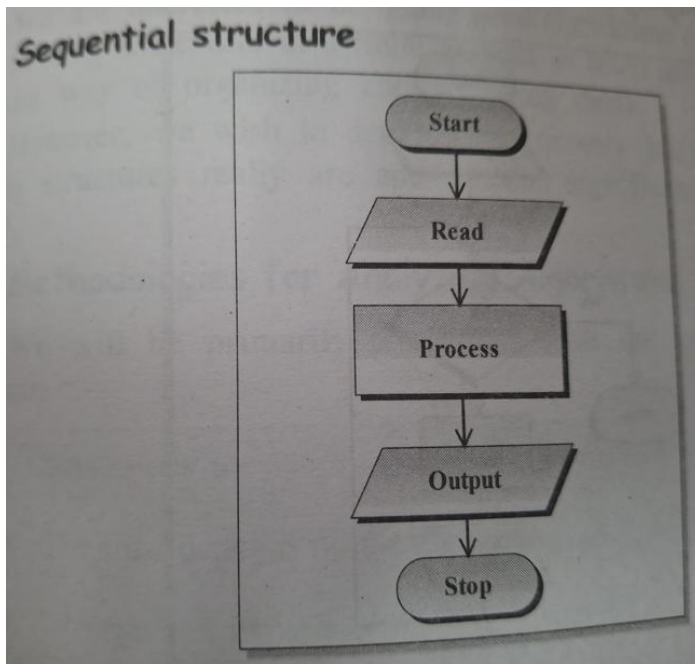
decision

flow of process →

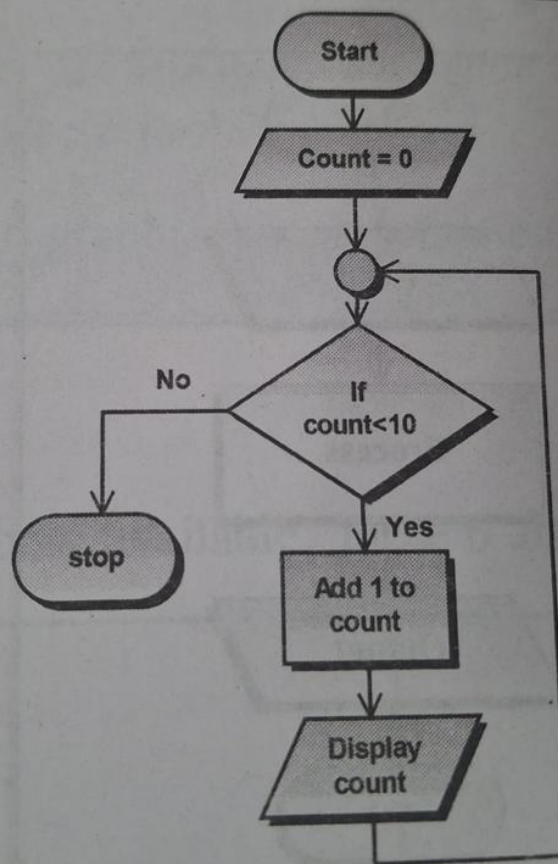
Basic Symbols

Of the many Symbols available, these 6 *Basic Symbols* will be used most;

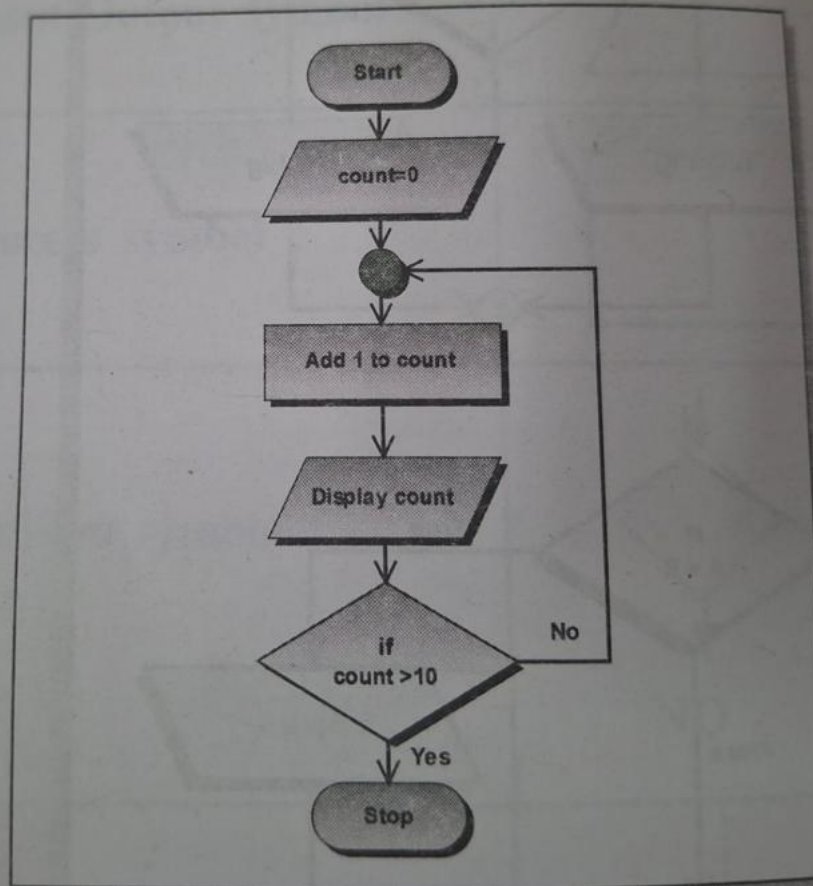
Flowlines :	
Terminal symbol :	
Input/Output symbol :	
Process symbol :	
Decision symbol :	
Connectors:	



Repetition/Looping Structure



Combination of Structures



Write an algorithm to calculate sum of two numbers

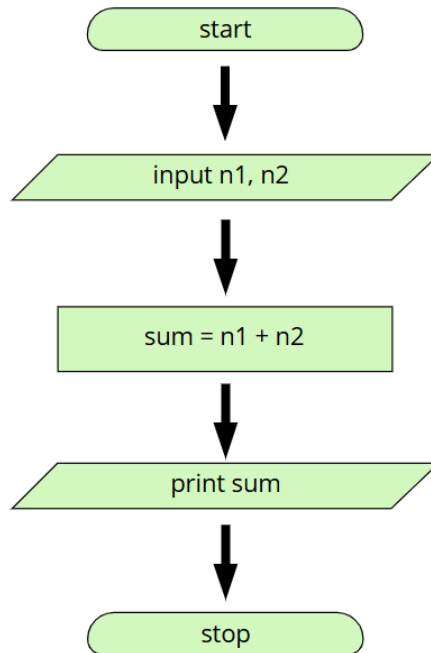
Algorithm

```
step 1 : start
step 2 : read n1, n2
step 3 : sum = n1 + n2
step 4 : print sum
step 5 : stop
```

Pseudo code

START
INPUT n1, n2
sum $\leftarrow$ n1 + n2
PRINT sum
STOP

Flowchart



Write an algorithm to calculate area of rectangle.

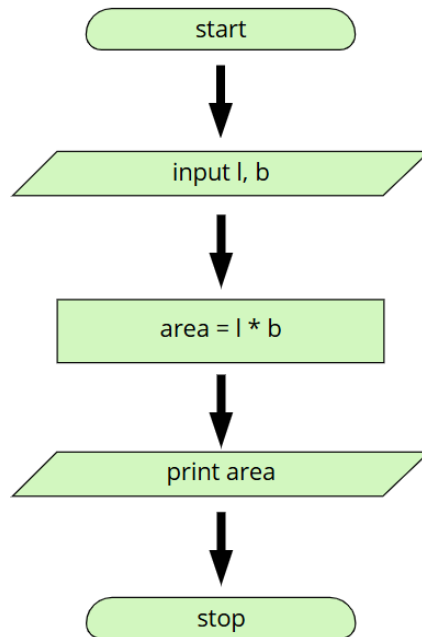
Algorithm

step 1 : start
step 2 : read l, b
step 3 : area = l * b
step 4 : print area
step 5 : stop

Pseudo code

START
INPUT l, b
area $\leftarrow$ l * b
PRINT area
STOP

Flowchart



Swapping of two number

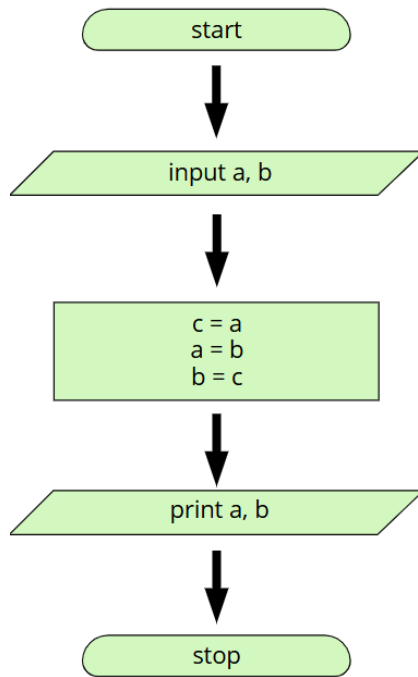
Algorithm

Step 1 : start
step 2 : read a & b
step 3 : c = a
 a = b
 b = c
step 4 : print a and b
step 5 : stop

Pseudo code

START
INPUT a, b
c ← a
a ← b
b ← c
PRINT a, b
STOP

Flowchart



Check whether any positive number is single digit or multi digit.

Algorithm

```

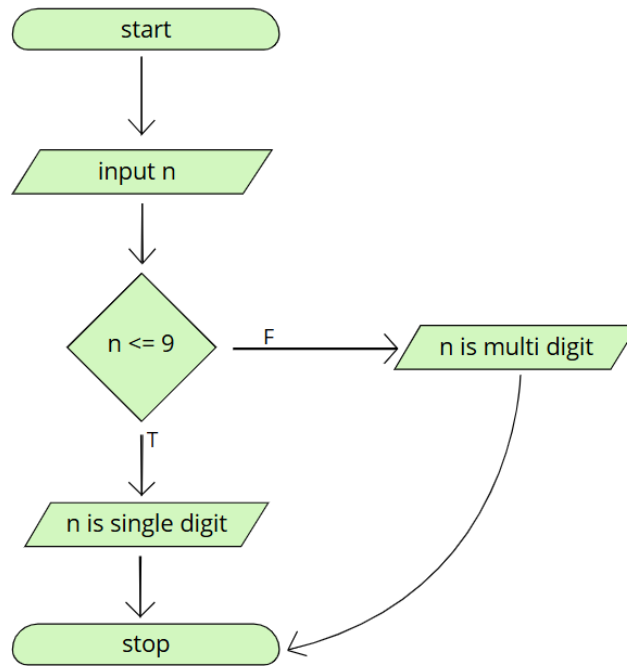
Step 1 : start
Step 2 : read n
Step 3 : if  $n \leq 9$ 
print number is single digit
else
print number is multi digit
Step 4 : stop
  
```

Pseudo code

```

START
INPUT n
IF ( $n \leq 9$ ) THEN
  PRINT "Number is single digit"
ELSE
  PRINT "Number is multi digit"
END IF
STOP
  
```

Flowchart



Print first n natural numbers

Algorithm

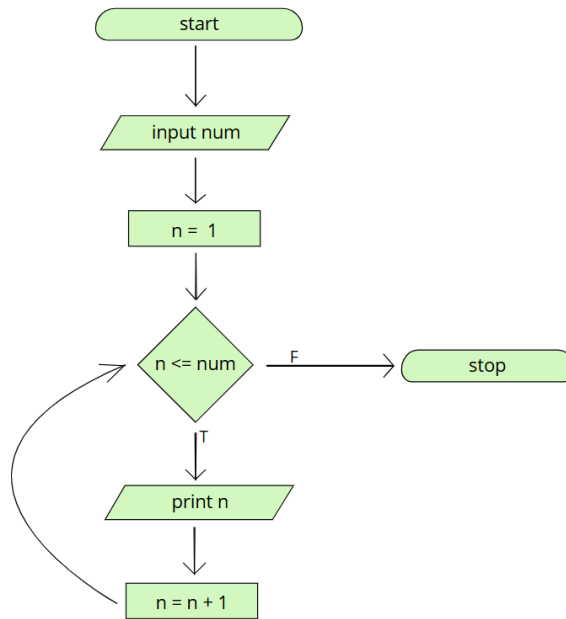
Step 1 : start
 Step 2 : read num
 Step 3 : $n = 1$
 Step 4 : If $n \leq \text{num}$, go to Step 5; otherwise go to Step 6
 Step 5 : print n;
 $n = n + 1$; go to Step 4
 Step 6 : stop

Pseudo code

```

START
INPUT num
 $n \leftarrow 1$ 
WHILE ( $n \leq \text{num}$ ) DO
  PRINT n
   $n \leftarrow n + 1$ 
END WHILE
STOP
  
```

Flowchart



Input 4 marks of subject and calculate percentage

and make result according to following

percentage $\geq 70 \rightarrow$ 1st division with honours

$60 \leq \text{percentage} \leq 70 \rightarrow$ 1st division

$50 \leq \text{percentage} \leq 60 \rightarrow$ second division

$40 \leq \text{percentage} \leq 50 \rightarrow$ third division

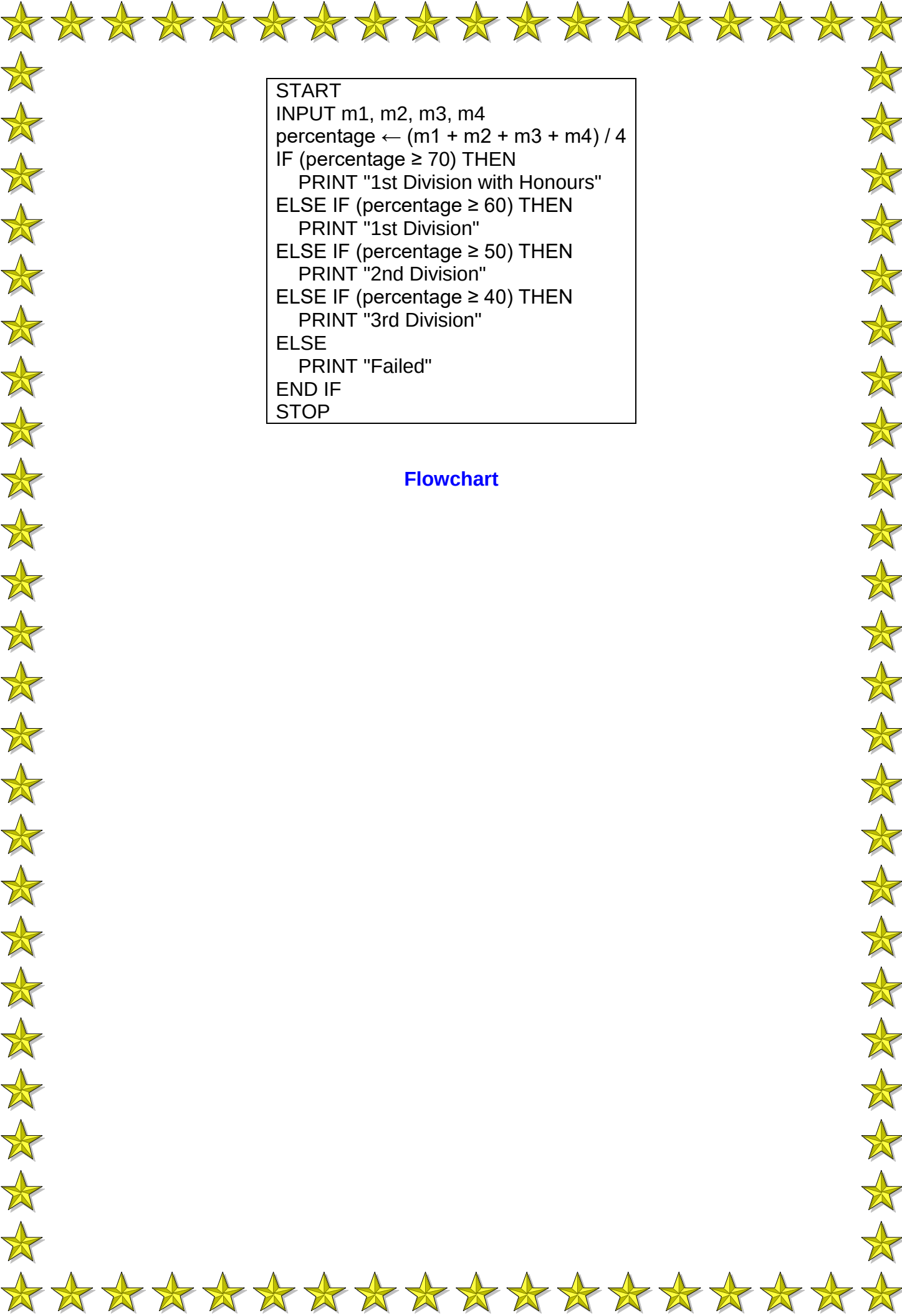
percentage $< 40 \rightarrow$ failed

Algorithm

```

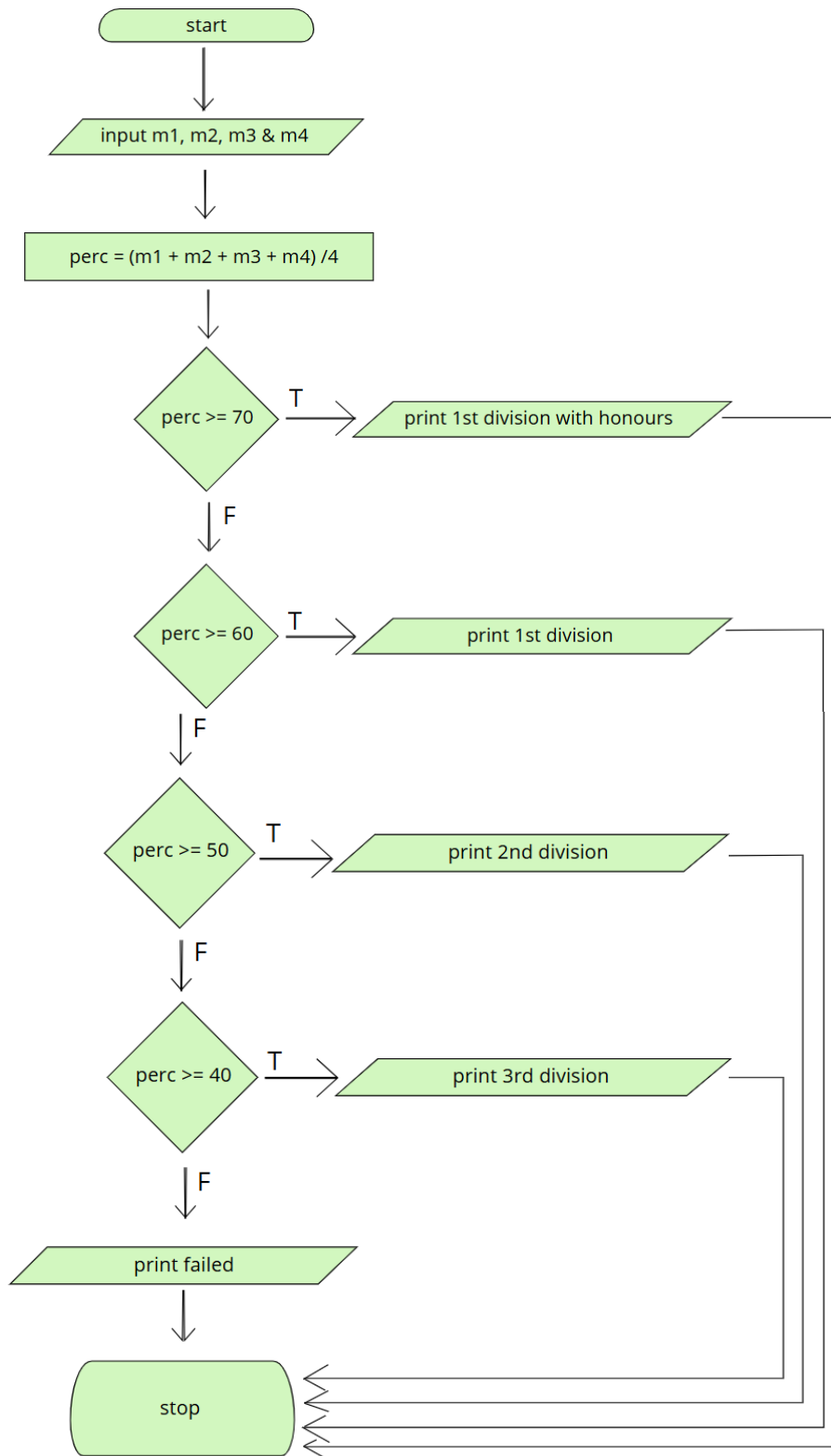
Step 1 : start
Step 2 : read m1, m2, m3 & m4
Step 3 : percentage = (m1 + m2 + m3 + m4)/4
Step 4 : if (percentage >= 70)
print 1st division with honours
else if (percentage >= 60)
print 1st division
else if (percentage >= 50)
print 2nd division
else if (percentage >= 40)
print 3rd division
else
print failed
Step 5 : stop
  
```

Pseudo code



```
START
INPUT m1, m2, m3, m4
percentage  $\leftarrow (m1 + m2 + m3 + m4) / 4$ 
IF (percentage  $\geq 70$ ) THEN
    PRINT "1st Division with Honours"
ELSE IF (percentage  $\geq 60$ ) THEN
    PRINT "1st Division"
ELSE IF (percentage  $\geq 50$ ) THEN
    PRINT "2nd Division"
ELSE IF (percentage  $\geq 40$ ) THEN
    PRINT "3rd Division"
ELSE
    PRINT "Failed"
END IF
STOP
```

Flowchart



Compute the largest of given three numbers

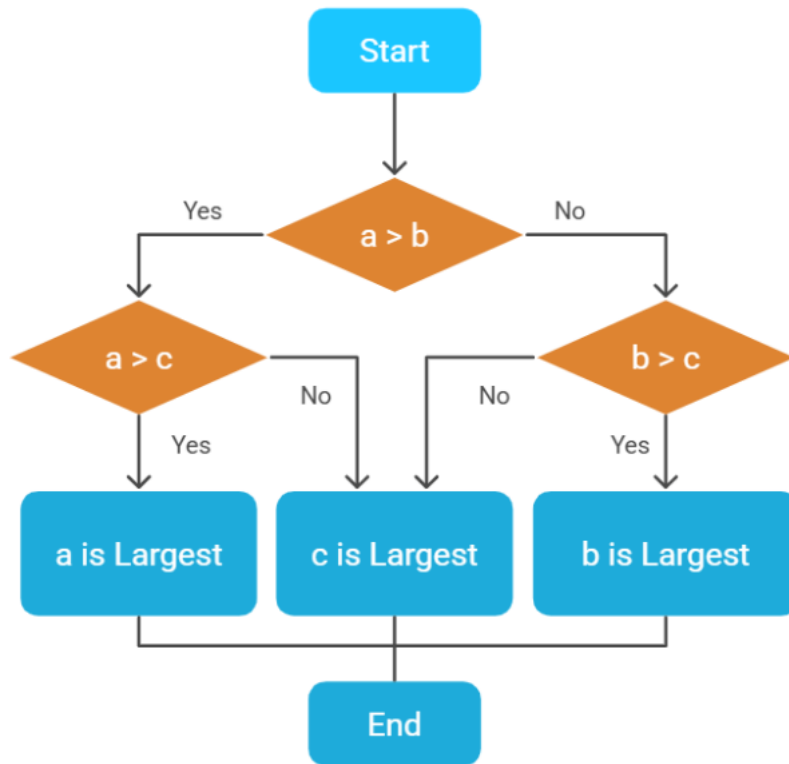
Algorithm

Step 1. Start
Step 2. Read A, B, C
Step 3. Compare A with B
Step 4. If $A > B$, go to Step 5; otherwise, go to Step 6
Step 5. If $A > C$, print A as the largest; otherwise, print C as the largest; go to Step 8
Step 6. compare B with C
Step 7. If $B > C$, print B as the largest; otherwise, print C as the largest
Step 8. Stop

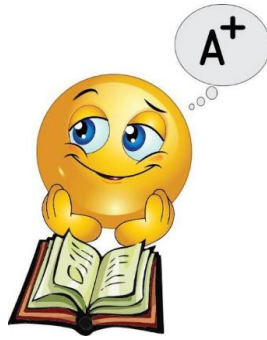
Pseudo code

```
START  
  
INPUT A, B, C  
  
IF A > B THEN  
    IF A > C THEN  
        PRINT "A is largest"  
    ELSE  
        PRINT "C is largest"  
    END IF  
ELSE  
    IF B > C THEN  
        PRINT "B is largest"  
    ELSE  
        PRINT "C is largest"  
    END IF  
END IF  
  
STOP
```

Flowchart



Algorithm	Pseudocode
A step-by-step procedure to solve a problem	A structured, programming-like representation of the algorithm
Focuses on what steps to perform	Focuses on how to organize those steps like a program
Usually uses step numbers	Usually uses programming constructs
Can explicitly say "go to Step 5"	Uses IF, ELSE, WHILE, FOR, etc.
Does not require programming-style syntax	Looks similar to a programming language, but is not actual code
Example: If A > B, go to Step 4	Example: IF A > B THEN ... ELSE ... END IF
Example: 1. If $i \leq n$, go to Step 2, otherwise go to Step 4 2. $\text{sum} \leftarrow \text{sum} + i$ 3. $i \leftarrow i + 1$, go to Step 1 4. Print sum	Example: WHILE $i \leq n$ DO $\text{sum} \leftarrow \text{sum} + i$ $i \leftarrow i + 1$ END WHILE PRINT sum



1. Create an algorithm, pseudo code and a flowchart to find the smallest of three numbers
2. Create an algorithm, pseudo code and a flowchart that will output for GCD of 2 numbers.
3. Create an algorithm, pseudo code and a flowchart that will output whether the given number is prime or not.